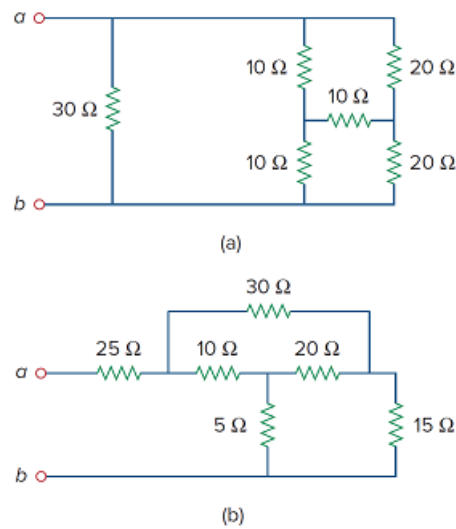


Problem

Obtain the equivalent resistance at the terminals $a-b$ for each of the circuits in Fig. 2.115.

Figure 2.115:



Step-by-step solution

Step 1 of 9

(a)

Refer to Figure 2.115 (a) in the textbook.

Draw the resistive circuit with notations.

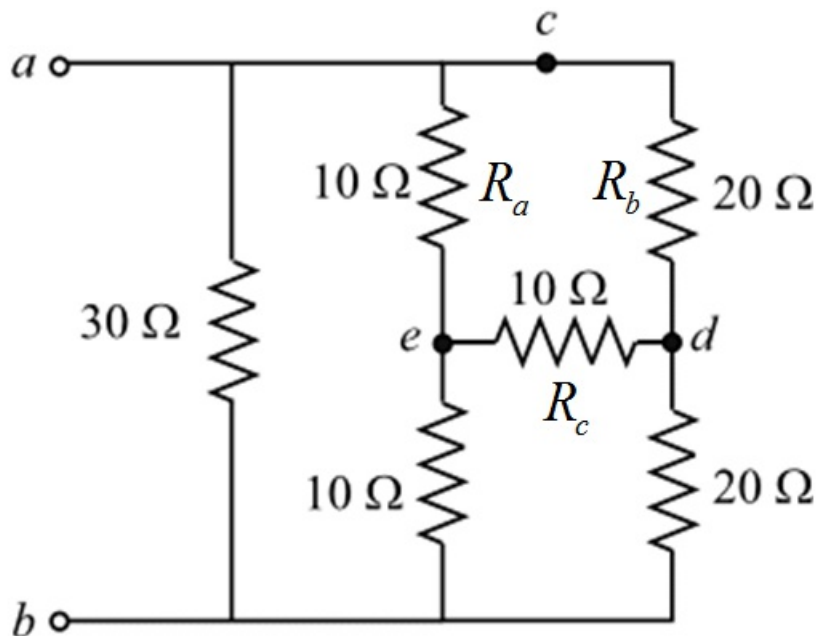


Figure 1

Step 2 of 9

Convert the top delta connected resistances marked as *cde* to star connection.

Redraw the circuit in Figure 1 by replacing the delta by star network.

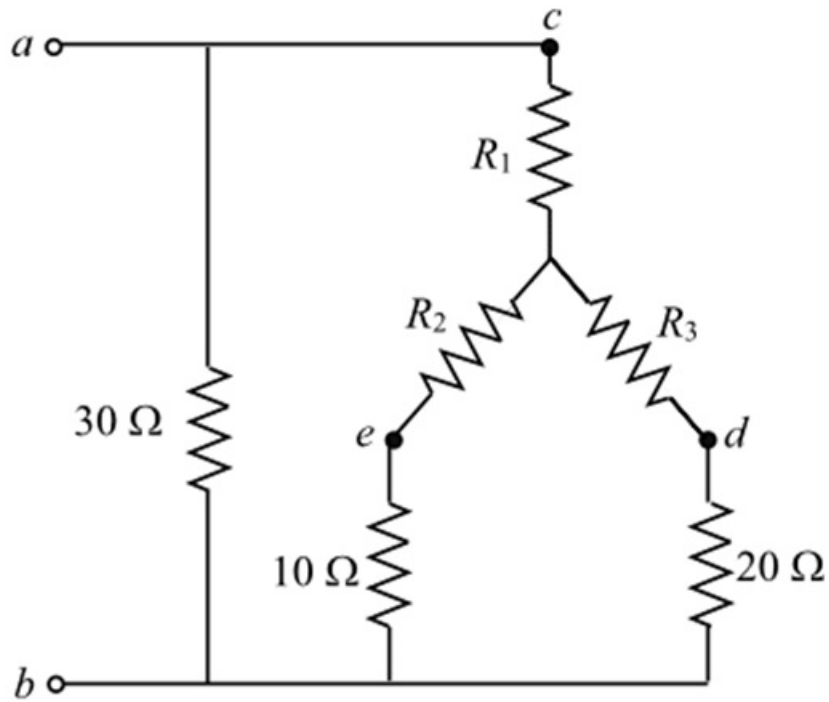


Figure 2

Step 3 of 9

Calculate the value of R_1

$$\begin{aligned} R_1 &= \frac{R_a R_b}{R_a + R_b + R_c} \\ &= \frac{(10)(20)}{10 + 20 + 10} \\ &= \frac{(10)(20)}{40} \\ &= 5\ \Omega \end{aligned}$$

Calculate the value of R_2

$$\begin{aligned} R_2 &= \frac{R_a R_c}{R_a + R_b + R_c} \\ &= \frac{(10)(10)}{10 + 20 + 10} \\ &= \frac{(10)(10)}{40} \\ &= 2.5\ \Omega \end{aligned}$$

Step 4 of 9

Calculate the value of R_3 .

$$\begin{aligned} R_3 &= \frac{R_b R_c}{R_a + R_b + R_c} \\ &= \frac{(10)(20)}{10 + 20 + 10} \\ &= \frac{(10)(20)}{40} \\ &= 5 \, \Omega \end{aligned}$$

Calculate the equivalent resistance of the branch, R_{cb} .

$$\begin{aligned} R_{cb} &= R_1 + (R_2 + 10) \parallel (R_3 + 20) \\ &= 5 + (2.5 + 10) \parallel (5 + 20) \\ &= 5 + (12.5 \parallel 25) \\ &= 5 + \frac{(12.5)(25)}{12.5 + 25} \\ &= 5 + 8.3333 \\ &= 13.3333 \, \Omega \end{aligned}$$

Step 5 of 9

Calculate the equivalent resistance of the network, R_{ab}

$$R_{ab} = 30 \parallel R_{cb}$$

Substitute 13.3333 for R_{cb} .

$$\begin{aligned} R_{ab} &= 30 \parallel 13.33 \\ &= \frac{(13.3333)(30)}{13.3333 + 30} \\ &= 9.231 \, \Omega \end{aligned}$$

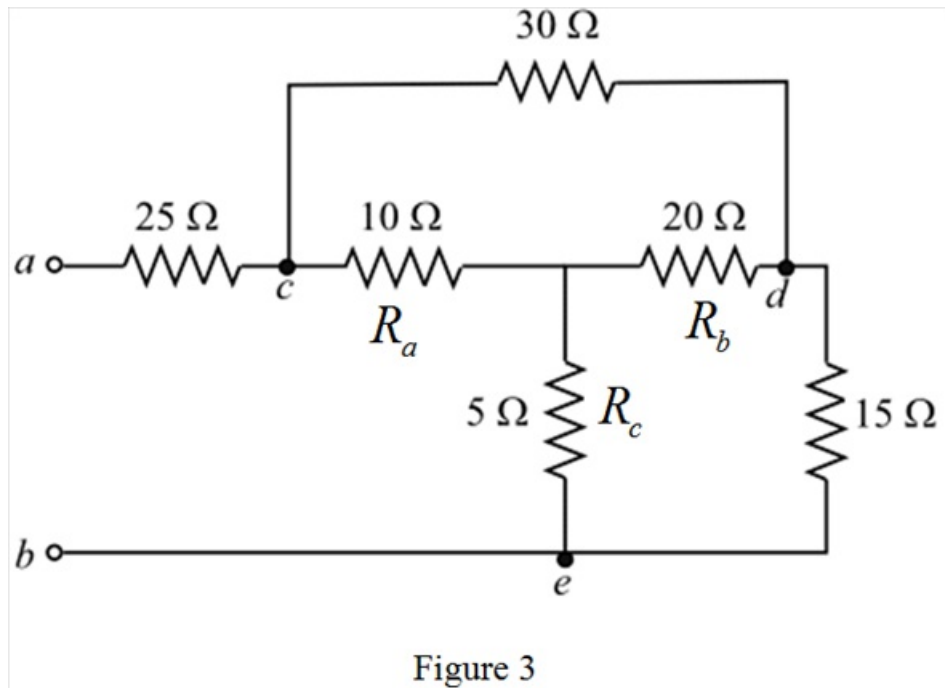
Thus, the equivalent resistance of the network at the terminals a - b is 9.231 Ω .

Step 6 of 9

(b)

Refer to Figure 2.115 (a) in the textbook.

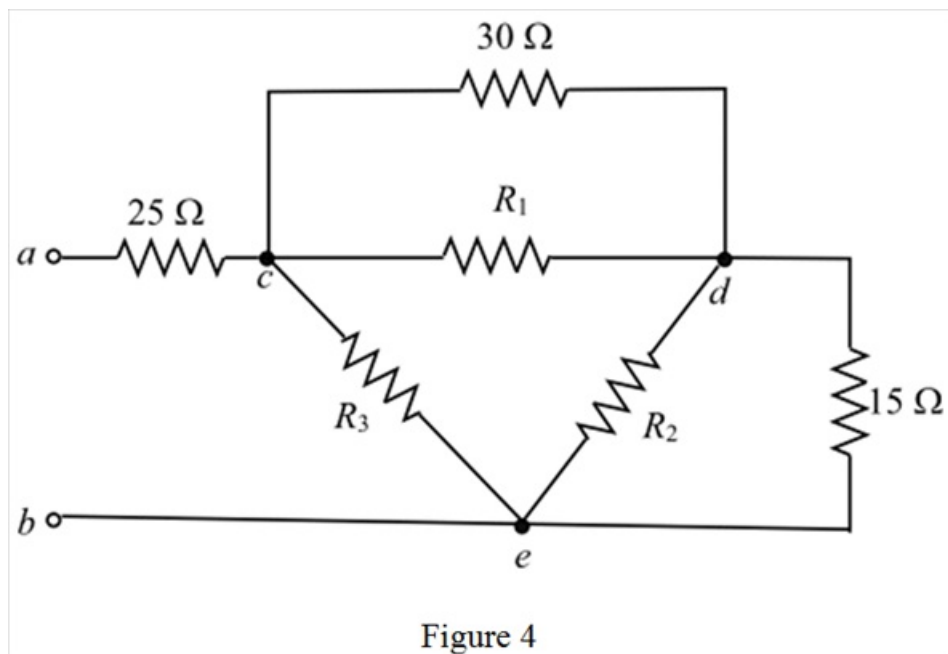
Draw the resistive circuit with notations.



Step 7 of 9

Convert the star cde to delta network.

Redraw the circuit in Figure 3 by transforming star to delta connection, as shown in Figure 4.



Step 8 of 9

Calculate the value of R_1 .

$$\begin{aligned} R_1 &= \frac{R_a R_b + R_b R_c + R_c R_a}{R_c} \\ &= \frac{(10)(20) + (10)(5) + (20)(5)}{5} \\ &= \frac{350}{5} \\ &= 70 \, \Omega \end{aligned}$$

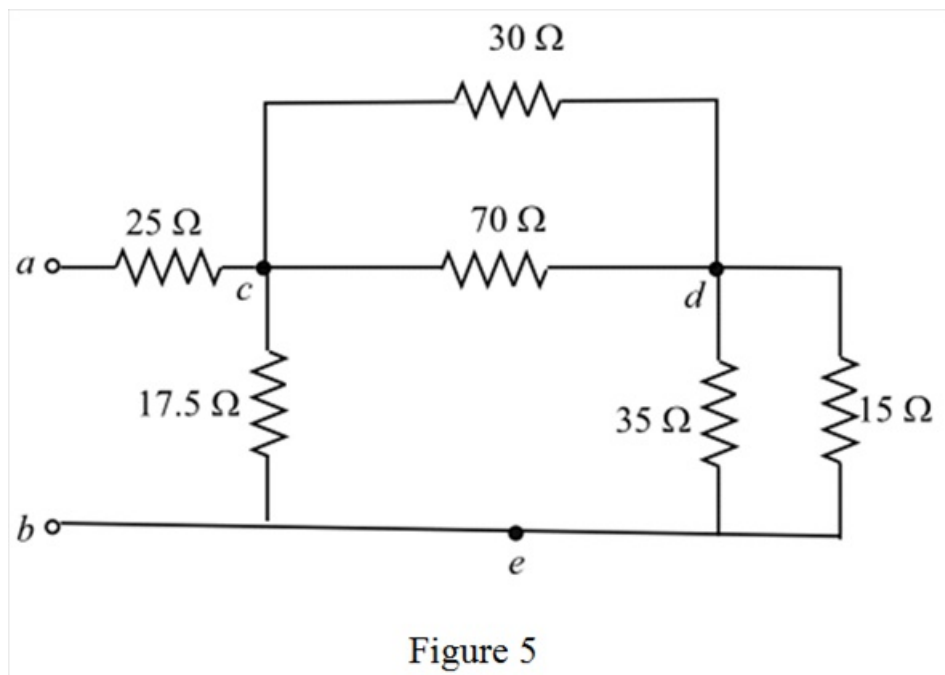
Calculate the value of R_2 .

$$\begin{aligned} R_2 &= \frac{R_a R_b + R_b R_c + R_c R_a}{R_a} \\ &= \frac{(10)(20) + (10)(5) + (20)(5)}{10} \\ &= \frac{350}{10} \\ &= 35 \, \Omega \end{aligned}$$

Calculate the value of R_3 .

$$\begin{aligned} R_3 &= \frac{R_a R_b + R_b R_c + R_c R_a}{R_b} \\ &= \frac{(10)(20) + (10)(5) + (20)(5)}{20} \\ &= \frac{350}{20} \\ &= 17.5 \, \Omega \end{aligned}$$

Redraw the circuit with the resistances marked as shown in Figure 5.



Calculate the equivalent resistance, R_{ab} .

$$\begin{aligned} R_{ab} &= 25 + \left[17.5 \parallel \left\{ (30 \parallel 70) + (35 \parallel 15) \right\} \right] \\ &= 25 + \left[17.5 \parallel \left\{ \left(\frac{30 \times 70}{30 + 70} \right) + \left(\frac{35 \times 15}{35 + 15} \right) \right\} \right] \\ &= 25 + \left[17.5 \parallel \left\{ (21) + (10.5) \right\} \right] \\ &= 25 + (17.5 \parallel 31.5) \\ &= 25 + \left(\frac{17.5 \times 31.5}{17.5 + 31.5} \right) \\ &= 25 + 11.25 \\ &= 36.25 \, \Omega \end{aligned}$$

Thus, the equivalent resistance of the network at the terminals $a-b$ is $36.25 \, \Omega$.